

Histidine-rich glycoprotein: the Swiss Army knife of mammalian plasma

Ivan K. H. Poon,^{1,2} Kruti K. Patel,² David S. Davis,¹ *Christopher R. Parish,¹ and *Mark D. Hulett^{1,2}

¹The John Curtin School of Medical Research, Australian National University, Canberra, Australia; and ²Department of Biochemistry, La Trobe Institute for Molecular Science, La Trobe University, Melbourne, Australia

Histidine-rich glycoprotein (HRG), also known as histidine-proline-rich glycoprotein, is an abundant and well-characterized protein of vertebrate plasma. HRG has a multidomain structure that allows the molecule to interact with many ligands, including heparin, phospholipids, plasminogen, fibrinogen, immunoglobulin G, C1q, heme, and Zn²⁺. The ability of

HRG to interact with various ligands simultaneously has suggested that HRG can function as an adaptor molecule and regulate numerous important biologic processes, such as immune complex/necrotic cell/pathogen clearance, cell adhesion, angiogenesis, coagulation, and fibrinolysis. The present review covers the proposed multifunctional roles of HRG

with a focus on recent findings that have led to its emergence as a key regulator of immunity and vascular biology. Also included is a discussion of the striking functional similarities between HRG and other important multifunctional proteins found in plasma, such as C-reactive protein, C1q, β_2 glycoprotein I, and thrombospondin-1. (*Blood*. 2011;117(7):2093-2101)

Introduction

Histidine-rich glycoprotein (HRG) is a ~ 75-kDa single polypeptide chain protein. HRG was first isolated and characterized from human serum in 1972^{1,2} and later found to be present in the plasma of many vertebrates³ and in aquatic invertebrates.⁴ Human HRG is synthesized in the liver⁵ and is present in plasma at the relatively high concentration of ~ 100-150 μ g/mL (~ 1.5 μ M). HRG has also been detected on the surface of leukocytes such as macrophages and monocytes,⁶ as well as in the α -granules of platelets and megakaryocytes.⁷

The human *HRG* gene has been mapped to position 3q28-q29 on chromosome 3⁸ and is predicted to encode a 507-amino acid⁵ multidomain protein consisting of 2 N-terminal regions with homology to cystatin-like domains (termed N1 and N2), a central histidine-rich region (HRR) flanked by 2 proline-rich regions (PRR1 and PRR2), and a C-terminal domain, or C (Figure 1). Although HRG has been classified as a member of the cystatin supergene-family that are generally known as cysteine protease inhibitors,⁹ no protease inhibitory activity has been reported for HRG. HRG also contains 4 intradomain and 2 interdomain disulfide bridges,¹⁰ and 6 predicted N-linked glycosylation sites.^{2,5} The protein and gene structure of HRG have been reviewed in detail by Jones et al.³

A variety of molecules have been shown to interact with HRG, including heme,¹¹ Zn²⁺,¹² plasminogen,¹³ heparanase,¹⁴ fibrinogen,¹⁵ thrombospondin (TSP),¹⁶ vasculostatin,¹⁷ immunoglobulin G (IgG),¹⁸ complement components,^{18,19} and heparin.¹ HRG can also interact with cell-associated molecules, including Fc γ receptors (Fc γ R),²⁰ heparan sulfate (HS),²¹ phospholipids,²² tropomyosin,²³ ATP synthase,²⁴ DNA,²⁵ and cytoplasmic ligand(s) exposed on necrotic cells.²⁶ HRG domains and the predicted ligand binding sites are depicted in Figure 1. On the basis of the modular architecture of HRG, it has been proposed that HRG may act as an adaptor molecule that interacts with multiple ligands simultaneously through several independent binding sites.²⁷ Collectively, HRG can potentially regulate numerous biologic processes, with

the first section of this review focusing on the role of HRG in immunity and vascular biology.

The role of HRG in immunity

HRG modulates the formation of immune complexes

The formation of immune complexes (ICs) plays an important role in a normal immune response, whereby specific antibodies form complexes with target antigen and facilitate the clearance and neutralization of microorganisms or foreign substances. However, if ICs are not cleared properly from the circulation, their depositions at target tissues could result in diseases such as arthritis, vasculitis, and glomerulonephritis. Besides components of the complement system, HRG has been shown to interact with IgG and ICs, subsequently regulating the clearance and formation of insoluble IC (IIC).^{18-20,28,29}

Human HRG was first found to interact with both rabbit and human IgG¹⁸ and later shown to bind different IgG subclasses and IgG molecules containing κ and λ light chains with different affinities.²⁹ HRG binding to IgG is dependent on its N-terminal domain and possibly involves HRG interacting with the F(ab) region of IgG.¹⁸ Significantly, the HRG-IgG interaction seems to play an important role in regulating IIC formation and clearance. Studies by Gorgani et al¹⁸ and Manderson et al¹⁹ demonstrated that HRG is an important component in human plasma that is able to inhibit the formation of IIC. Furthermore, HRG can prevent the formation of IIC generated by rheumatoid factor (an auto-anti-Fc antibody) and human IgG, possibly by masking the epitopes on IgG recognized by rheumatoid factor.²⁸ HRG can also aid the solubilization of IIC,²⁸ enhance complement activation by IIC,¹⁹ as well as modulate the binding of IC to monocytes.²⁰

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*C.R.P. and M.D.H. contributed equally to this work.

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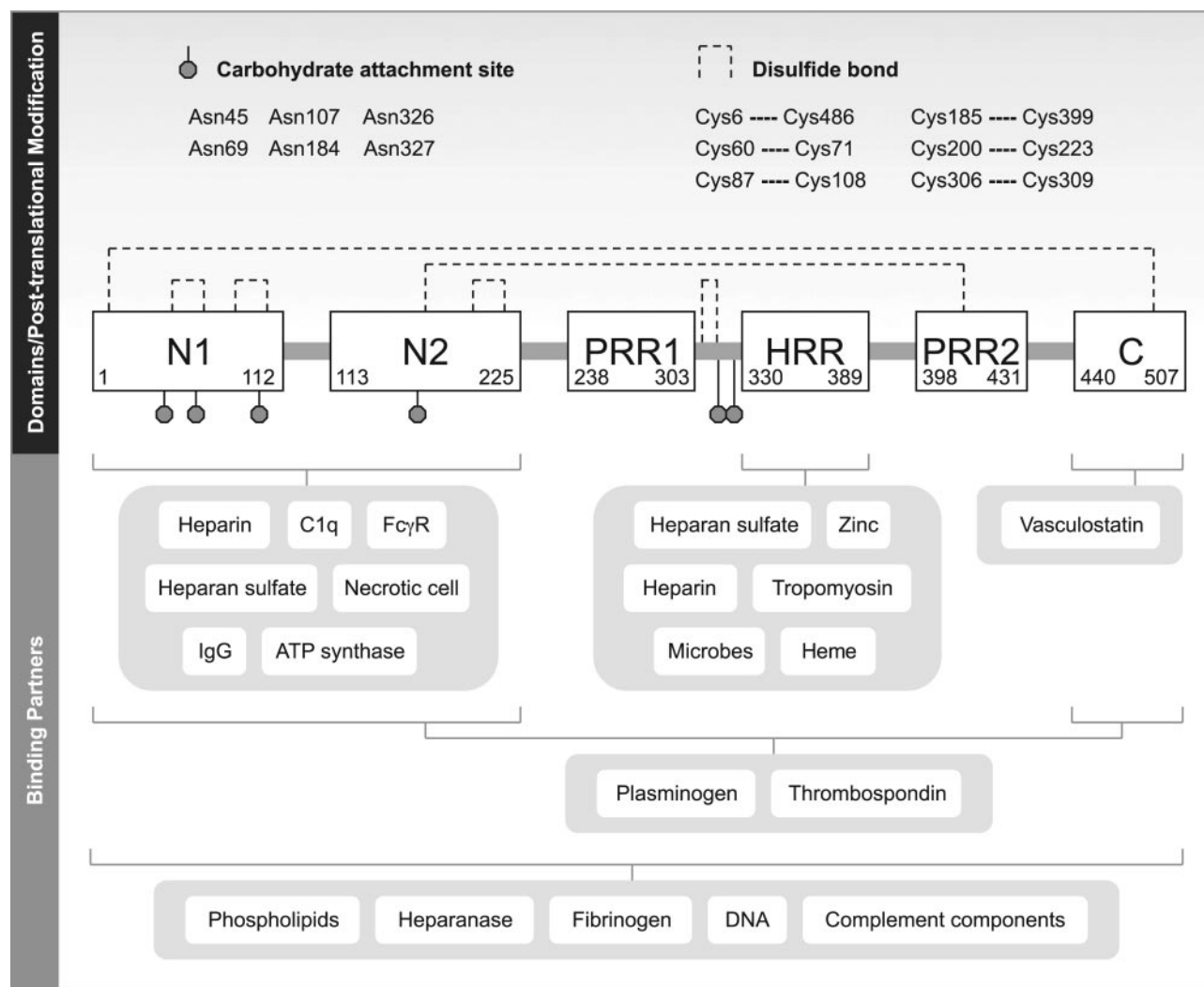


Figure 1. Schematic representation of protein domains, posttranslational modifications, and proposed binding partners of human HRG. HRG is a multidomain protein consisting of 2 cystatin-like regions at the N-terminus (N1 and N2), a central histidine-rich region (HRR), 2 proline-rich regions (PRR1 and PRR2) flanking the HRR, and a C-terminal domain (C). Proposed carbohydrate attachment sites, disulfide bridges, binding partners of HRG, and the corresponding binding regions of HRG (ie, N1N2, HRR, or C) are depicted in the figure. Note that the binding sites for phospholipids, heparanase, fibrinogen, DNA, and certain complement components on HRG have not been characterized.

HRG regulates FcγR function

Leukocyte Fc receptors play a vital role in linking the humoral and cellular arms of the immune system by aiding the interaction of antigen-specific antibodies with nonspecific effector cells that express Fc receptors. In humans, there are 3 classes of IgG binding FcγR, namely FcγRI, FcγRII, and FcγRIII. These FcγR are expressed at differential levels on a variety of cell types and regulate diverse biologic processes. Interestingly, HRG has been shown to regulate the expression and functions of FcγR.^{20,30,31}

Initially, HRG was demonstrated to modulate FcγR (mainly FcγRII)-mediated phagocytosis of IgG-opsonized sheep erythrocyte in a biphasic manner depending on the duration of pretreatment of macrophages with HRG.³¹ Later studies by Chang et al³⁰ suggested that the effects of HRG on FcγR-dependent phagocytosis is attributed to the regulation of FcγRII expression and protein synthesis by HRG. Furthermore, HRG was suggested to regulate monomeric IgG binding to FcγRI through direct interaction with FcγRI.²⁰ Although the *in vivo* role of HRG in modulating FcγR function is unclear, investigators have demonstrated the importance of HRG in aiding the clearance of IC²⁰ and dying/dead cells^{22,25} via

an FcγR-dependent mechanism. However, it is important to note that the presence of a small amount of IgG in HRG preparations purified from human plasma²² raises doubts about the validity of studies in which the authors demonstrated a direct interaction of HRG with FcγR.

HRG facilitates the removal of dying/dead cells

Under normal physiologic conditions, rapid removal of dying/dead cells (eg, apoptotic and necrotic cells) from the circulation and tissues by phagocytic cells plays a critical role in maintaining tissue homeostasis and turnover. Impaired dying/dead cell clearance could result in the exposure of antigenic intracellular molecules, which can lead to the development of autoimmune diseases such as systemic lupus erythematosus.³² HRG was shown to facilitate the removal of late (ie, plasma membrane permeabilized) apoptotic²⁵ and necrotic cells.^{22,26,33}

Studies by Gorgani et al²⁵ showed that HRG binds strongly to late apoptotic cells compared with viable or early (ie, plasma membrane intact) apoptotic cells, possibly by recognizing naked DNA. Consistent with previous studies,²⁰ HRG was suggested to

bind Fc γ RI on macrophages and function as a bridging molecule to augment the uptake of late apoptotic cells via an Fc γ RI-dependent mechanism.²⁵ The authors also demonstrated that, like components of the complement system,³⁴ HRG is involved in normal human serum-dependent phagocytosis of late apoptotic cells.²⁵ Similarly, studies by Jones et al²⁶ demonstrated that, in addition to cell surface HS, HRG binds strongly to cytoplasmic ligand(s) exposed on permeabilized necrotic cells via its N1N2 domain. Recently, the molecular mechanisms underpinning enhanced necrotic cell uptake by HRG were characterized in detail, with HRG functioning as a pattern recognition molecule (PRM) that binds various intracellular phospholipids exposed on necrotic cells and acts as an adaptor protein to recruit IgG and aid the clearance of necrotic cells via Fc γ RI and HS on phagocytes.^{22,33} Furthermore, HRG has been reported to interact with C1q¹⁸ and TSP,^{16,35,36} both of which are also involved in the uptake of apoptotic and necrotic cells.³² Therefore, these opsonins may work in concert to potentiate the removal of dying/dead cells and determine the subsequent immune response. Indeed, recent studies by Manderson et al¹⁹ demonstrated that HRG can regulate the activation of complement on necrotic cells, possibly via direct interaction with complement components such as C1q, factor H and C4b-binding protein.

HRG exhibits antimicrobial and endotoxin-neutralizing properties

The innate immune system often uses PRMs (eg, CD14, C-reactive protein [CRP], C1q, and mannose binding lectin) that recognize conserved molecular patterns on pathogens or dying/dead cells and aid their clearance by phagocytes.³² In addition to dying/dead cell removal, HRG and peptides derived from the HRR of HRG have been shown to exert antimicrobial activities.³⁷⁻⁴⁰

Similar to other histidine-rich and heparin-binding peptides, peptides derived from the HRR of HRG, such as the 20mer peptide (GHHPH)₄, were able to bind to heparin³⁷ and exert antimicrobial activity against the Gram-positive bacteria *Enterococcus faecalis*, the Gram-negative bacteria *Escherichia coli*,³⁷ and the fungus *Candida albicans*^{39,40} under acidic conditions or in the presence of Zn²⁺. Furthermore, unlike the closely related cystatin superfamily member high-molecular-weight kininogen (HMWK), which requires proteolytic cleavage by elastase to generate antimicrobial fragments/peptides,⁴¹ intact plasma-derived and recombinant HRG were able to exert antibacterial³⁷ and antifungal effects.⁴⁰ In the presence of Zn²⁺ or at low pH, intact HRG was able to bind and trigger membrane destabilization, as well as the release of the cytoplasmic contents of whole bacteria and fungi. However, the HRR seems to be essential for this process as recombinant HRG containing only the 2 N-terminal cystatin-like regions showed no antimicrobial activity.^{37,40} Recently, the antimicrobial role of HRG was validated in vivo by the use of HRG-deficient (HRG^{-/-}) mice, with HRG^{-/-} mice being found to be more susceptible to *Candida albicans* and *Streptococcus pyogenes* infections than their wild-type counterparts (Table 1).^{6,17-20,22,25,26,28,33,38,40,42-50}

The cationicity of many antimicrobial peptides is known to be critical for the initial electrostatic attraction of peptides to the highly electronegative surface of bacteria and fungi, as well as the ability of peptides to traverse the microbial plasma membrane to aid pathogen killing via either membrane- or nonmembrane-disruptive mechanisms. Although no bacterial ligand(s) of HRG have yet been identified, one could speculate that the cationicity of the HRR may play an important role because the antimicrobial effects of HRG and the HRR-derived peptides were highly dependent on low pH or the presence of Zn²⁺,^{37,40} both of which

impose a net positive charge on the molecule. Interestingly, a synthetic peptide containing the histidine-rich consensus sequence of HRG was able to neutralize the lipopolysaccharide (LPS)-induced proinflammatory response by dampening LPS-induced IL-8 production by CD14-transfected THP-1 cells.⁴² Despite the lack of direct evidence of an interaction between HRG and the anionic bacterial component LPS, HRG may act as a PRM that recognizes anionic molecules, most likely anionic phospholipids, exposed on microorganisms and facilitate their removal either via direct killing and/or phagocytosis in an analogous fashion to the recently described recognition and clearance of necrotic cells by HRG.²²

HRG modulates cell adhesion

Immune cells use a variety of cell adhesion molecules to mediate communication with other cells as well as migration to inflammatory sites. The authors of previous studies have suggested that HRG may play either a positive or a negative role in regulating cell adhesion of immune cells such as murine T cells and macrophages.^{6,31,43-45}

In vitro studies with a T-cell line showed that the presence of HRG can enhance the adhesion and spreading of activated T cells on plastic tissue culture plates.⁴³ The combination of HRG and Zn²⁺ can also promote homotypic adhesion between T cells in culture.⁴⁴ Therefore, HRG may positively regulate cell adhesion. In contrast, other investigators have observed an inhibitory effect of HRG on the adhesion of immune cells, with HRG inhibiting the formation of autorosettes (cell adhesion between murine lymphocytes and autologous erythrocytes).^{6,45} Furthermore, studies by Olsen et al⁴⁴ showed that in the presence of Zn²⁺, HRG inhibits the adherence of a T-cell line to tissue culture plastic, as well as to extracellular matrix (ECM) components (eg, laminin, collagen, or fibronectin) coated to culture dishes. Similarly, Chang et al³¹ found that prolonged treatment of macrophages with HRG reduces their spreading and adherence to plastic wells. Although the molecular mechanisms underlying the diverse effects of HRG on cell adhesion are unclear, the authors of recent studies have suggested that HRG can directly interact with F-type ATP synthase on the surface of a T-cell line, possibly via the N1 domain, to promote cell adhesion and morphologic changes induced by concanavalin A.^{24,51} In addition, HRG and peptides derived from the HRR of HRG were demonstrated to modulate signal transduction events that are important in regulating cytoskeletal organization of endothelial cells,^{52,53} as well as interfering with $\alpha_v\beta_3$ integrin-mediated adhesion of endothelial cells to vitronectin.⁵²

The role of HRG in vascular biology

Regulation of angiogenesis by HRG

Angiogenesis, the formation of new blood vessels from preexisting vasculature, is essential in maintaining the delivery of sufficient oxygen and nutrients to all cells within an organism. Angiogenesis is tightly regulated by a multitude of endogenous activators and inhibitors, and involves numerous cell types. HRG and peptides derived from the HRR of HRG have been shown to exhibit both pro- and/or anti-angiogenic properties, depending on the experimental systems used.^{23,46,52-56}

One potential molecular mechanism underpinning the proangiogenic effects of HRG involves its strong interaction with TSP. TSP is a multifunctional homotrimeric α -granule glycoprotein that

Table 1. In vivo studies demonstrating a role for HRG in several physiological and pathological conditions

Properties	Experimental observations/future directions	Reference
Role of HRG in immunity		
Antifungal	Increased susceptibility to fungal infection in HRG-deficient mice At pH 5.5 (pathological pH of the wounded or infected foci) and in the presence of zinc, HRG exhibits antifungal activity in a concentration-dependant manner. The HRR tethers HRG to the ergosterol-containing cell wall, resulting in lysis of the target cell. HRG-deficient mice exposed to <i>Candida albicans</i> have a significantly greater rate of fatality, with exaggerated fungal load in lymphoid organs.	40
Antibacterial	HRG protects mice from <i>Streptococcus pyogenes</i> infection At pH 5.5, HRG lyses <i>Streptococcus pyogenes</i> via the HRR. The antifibrinolytic properties of HRG enable it to contain bacteria in a fibrin clot and its antibacterial activity may be exerted directly in the clot. HRG also orchestrates the containment of bacteria at the infection foci by modulating the chemotaxis of neutrophils/macrophages and the efficient formation of an abscess.	38
Endotoxin-neutralizing effects	In vitro data suggest that HRG-deficient mice will be more susceptible to septic shock.	42
Regulation of T-cell adhesion	In vitro data suggest that HRG-deficient mice will have an altered immune response as the result of a dysregulation in T-cell migration and interaction with antigen presenting cells/endothelium.	6,43-45
Apoptotic and necrotic cell clearance	In vitro data suggest that HRG-deficient mice (in particular aged mice) are likely to develop autoimmunity as the result of impairment in apoptotic and necrotic cell uptake. In vitro data also suggest that HRG-deficient mice may have a delayed response to necrotic cell-induced inflammation.	22,25,26,33
Formation of immune complexes	In vitro data suggest that HRG-deficient mice are likely to develop diseases such as arthritis, vasculitis, and glomerulonephritis as the result of an accumulation of insoluble immune complexes.	18-20,28
Role of HRG in vascular biology		
Proangiogenic (indirect)	HRG inhibits antiangiogenic activity of TSP-1 In both in vivo corneal angiogenesis assay and Matrigel plug assay on C57BL/6 mice, HRG reverses the inhibitory effect of TSP-1 on bFGF-induced neovascularization. Inhibition of vasculostatin by CLESH motif on HRG promotes brain tumor formation When coexisting at the tumor site, HRG abrogates the antiangiogenic activity of vasculostatin, resulting in enhanced glioblastoma volume in athymic nude mice implanted with LN229 glioma cells subcutaneously or intracranially. Histological analysis of mouse brain tumor specimen shows marked level of neovascularization in the presence of HRG.	17,46
Antiangiogenic	HRG inhibits vascularization and tumor establishment The tumor volume of subcutaneously inoculated T241 fibrosarcoma cells in C57BL/6 mice is significantly reduced by 60%-70% after HRG treatment. The presence of HRG increases the incidence of apoptosis and reduces cell proliferation at the tumor site. In a RCAS/TV-A in vivo mouse brain tumor model, HRG impairs growth of experimental glioblastoma with no noticeable effect on cell proliferation. By abrogating primary endothelial cell adhesion and rearrangement of focal adhesions in vitro, HRG may inhibit tumor vascularization via the same mechanism in vivo. The release of HRR fragments from HRG via proteolytic cleavage is essential for its antichemotactic properties in vivo. In a Rip1-Tag2 pancreatic β -cell tumor model, tumor angiogenesis is enhanced in HRG-deficient mice.	47-49
Anticoagulant and antifibrinolytic modifier	Enhanced blood coagulation and fibrinolysis in HRG-deficient mice The formation of blot clots is enhanced in the absence of plasma HRG because HRG-deficient mice exhibit shorter prothrombin time and shorter bleeding time. HRG is antifibrinolytic as the presence of HRG reduces the lysis of fibrin clots.	50

bFGF indicates basic fibroblast growth factor; HRG, histidine-rich glycoprotein; HRR, histidine-rich region; and TSP-1, thrombospondin-1.

inhibits angiogenesis through its interaction with the signaling receptor CD36, which delivers an antiangiogenic signal that blocks basic fibroblast growth factor (bFGF)-induced angiogenesis.⁴⁶ HRG was initially found to bind TSP with high affinity¹⁶ and was able to form a trimolecular complex with plasminogen and TSP.³⁵ Studies by Simantov et al⁴⁶ showed that HRG contains 2 CLESH motifs that resemble the TSP binding motif on CD36, and HRG was proposed to aid bFGF-induced angiogenesis by interfering with TSP-CD36-mediated antiangiogenic signaling.^{46,56} Similarly, HRG was shown recently to bind specifically to vasculostatin, the soluble antiangiogenic domain of brain angiogenesis inhibitor 1, via the CLESH motif on HRG and block the antiangiogenic effects

mediated through the vasculostatin-CD36 interaction.¹⁷ For both TSP and vasculostatin, their interaction with HRG is thought to be mediated via the type-1 repeats of TSP-1.^{17,46}

In contrast, HRG and peptides derived from the HRR of HRG were able to induce potent antiangiogenic effects both in vitro and in vivo via several mechanisms. HRG was shown to inhibit angiogenesis by modulating various signaling events that are important for endothelial cell survival, proliferation, and cell migration. For example, HRG was demonstrated to inhibit proliferation and induce apoptosis in endothelial cells in vitro via the activation of caspase-3.⁵⁵ Although studies by Olsson et al⁴⁷ also showed that tumors in HRG-treated mice have reduced

vascularization, increased apoptosis, and decreased proliferation (Table 1), HRG had no apparent effect on the survival or growth of endothelial cells *in vitro*. This effect was possibly attributable to an unlimited supply of oxygen and nutrients in *in vitro* cultures compared with the vasculature of tumor cells *in vivo*. Furthermore, HRG can directly transduce antiangiogenic signals by interacting with tropomyosin expressed on endothelial cells after bFGF activation.²³ HRG can also reduce chemotaxis of primary endothelial cells,⁴⁷ possibly caused by rearrangement of focal adhesion,^{47,52,53} disruption of cytoskeletal organization,^{52,53} inhibition of tube formation,⁵³ and reduction in cell attachment.^{47,52} The molecular mechanisms underlying the effects of HRG on the endothelial cell cytoskeleton was suggested to involve signaling events mediated through integrin-linked kinase⁵² and focal adhesion kinase,^{47,52,53} which could modulate processes that are important for angiogenesis such as lamellipodia formation, polarization and migration.⁵² Importantly, the antiangiogenic properties of HRG can be mediated solely via the HRR/PRR region⁵⁵ or peptides derived from the HRR of HRG.^{47,52-54} More recently, the presence of a proteolytic derived peptide corresponding to the HRR of HRG was identified in human tissues, supporting the notion that HRG can function as an endogenous regulator of angiogenesis.⁴⁸

The role of HRG in regulating angiogenesis is further complicated by its ability to modulate the activity of degradative enzymes such as the plasminogen/plasmin system that could in turn influence the formation of new vessels during angiogenesis. Extracellular proteases such as matrix metalloproteases and the plasminogen/plasmin system are critical in the remodeling of the ECM during vessel sprouting and the resolution phase of angiogenesis. Numerous investigators^{13,57-60} have demonstrated a strong interaction between HRG and plasminogen, which can either positively or negatively regulate the activation of plasminogen to the serine protease plasmin via plasminogen activators. Furthermore, HRG can regulate the function of another important ECM remodeling enzyme, the HS-degrading endo- β -D-glucuronidase, Heparanase, by both masking the cleavage sites of ECM HS⁶¹ and directly regulating the enzymatic activity of Heparanase.¹⁴

In addition, it is possible that HRG could exhibit pro- or antiangiogenic effects through controlling the availability of HS-binding growth factors such as FGF (an angiogenesis activator) to endothelial cells. Studies by Brown and Parish⁶² have demonstrated that HRG can compete with FGF for binding HS on cell surfaces, which results in inhibition of the mitogenic activity of FGF by preventing cell surface HS serving as a coreceptor for FGF. Alternatively, HRG could aid angiogenesis by displacing biologically active FGF sequestered by the HS component of the ECM.⁶² The ability of HRG to protect the Heparanase-sensitive areas of HS also suggests a potential role for HRG in either limiting angiogenesis by inhibiting Heparanase-mediated release of HS-binding growth factors from the ECM HS or by facilitating angiogenesis by preventing cleavage of endothelial cell surface HS, which serves as a coreceptor for FGF signaling.⁶¹ However, the observation that HRG can bind directly to Heparanase and potentiate its enzymatic activity¹⁴ adds another level of complexity to the regulation of the bioactivity of HS-binding growth factors by HRG. Moreover, the high-affinity interaction between HRG and heparin also limits the formation and activities of proangiogenic HMGB1-heparin and VEGF-A₁₆₅-heparin complexes.⁶³

Regulation of coagulation and fibrinolysis via HRG

Blood clotting (coagulation) and the dissolution of fibrin clots (fibrinolysis) are tightly regulated by a range of substrates,

activators, inhibitors, cofactors, and receptors to ensure precise prevention of blood loss and unnecessary blockage of vessels. During coagulation, exposure of damaged vascular surfaces results in the initial formation of a "platelet plug," which ultimately leads to the generation of thrombin and the formation of a thrombus through thrombin-mediated conversion of fibrinogen to fibrin, and by platelet activation. During fibrinolysis, the key fibrinolytic protease plasmin, is generated via proteolytic processing of the plasma zymogen plasminogen by tissue-type plasminogen activator and urokinase-type plasminogen activator. Interestingly, HRG has been shown to interact with components of both the coagulation and fibrinolytic systems, thus potentially playing an important role in regulating hemostasis.^{1,13,15,57,64,65}

Second to antithrombin III, HRG is one of the most abundant heparin-binding proteins in human plasma. HRG binds heparin with high affinity^{1,64} and was shown to neutralize the anticoagulant activity of heparin by preventing the formation of heparin-antithrombin III complexes that inhibit activated coagulation factors such as thrombin.^{64,66-68} Furthermore, the presence of Zn²⁺ that is released by activated platelets,⁶⁹ can potentiate the interaction between HRG and heparin,^{21,64} and the ability of HRG to neutralize heparin.^{70,71} Interestingly, HRG can also interact with fibrinogen and be incorporated into fibrin clots.¹⁵ Although it was observed that HRG had no effect on the extent of fibrinogen conversion into fibrin by thrombin during the formation of fibrin clots, the presence of HRG did retard the rate of conversion of fibrinogen to fibrin.¹⁵

As mentioned previously, HRG binds strongly to the lysine-binding site on plasminogen,¹³ possibly via its C-terminal lysine residues.^{57,60} Initially, HRG was proposed to be an antifibrinolytic agent by interfering with plasminogen interacting with binding partners that are important for its activation,¹³ such as fibrinogen, fibrin, integrin $\alpha_M\beta_2$, and annexin 2. It is interesting to note that HRG has been suggested to inhibit fibrinogen-dependent plasminogen activation⁵⁸ or have no apparent effect on fibrin-dependent plasminogen activation⁷² in solution. In contrast, HRG has also been suggested to function as a soluble plasminogen receptor that aids plasminogen activation by tethering plasminogen to glycosaminoglycans-coated surfaces⁵⁸ and cell surfaces.⁵⁹ Interestingly, recent studies by Poon et al⁶⁵ demonstrated that proteolytic cleavage of HRG by plasmin at specific sites may present a feedback mechanism to regulate the effects of HRG on the plasminogen/plasmin system.

The physiologic role of HRG in hemostasis was to some extent resolved by the generation of HRG^{-/-} mice by Tsuchida-Straeten et al.⁵⁰ This study suggested that HRG has both anticoagulation and antifibrinolytic properties *in vivo* (Table 1). HRG^{-/-} mice showed a significantly shorter plasma prothrombin time and shorter bleeding times than HRG^{+/+} mice, which indicates accelerated extrinsic clotting in HRG^{-/-} mice.⁵⁰ HRG^{-/-} mice also have enhanced fibrinolysis, whereby fibrin clots were lysed more rapidly in HRG^{-/-} mice compared with HRG^{+/+} mice.⁵⁰

The role of HRG in cancer progression

Tumorigenesis is known to be a multistep process that involves a variety of genetic alterations, which can progressively transform normal cells into aggressive malignant derivatives. During the process of transformation, cancer cells often hijack normal physiologic processes to facilitate growth and metastasis, such as the ability to induce angiogenesis to maintain tumor survival and to invade adjacent or distal tissues by modulating cell adhesion molecules and extracellular degradative enzymes. The ability of

Table 2. Functional similarities between HRG and other plasma proteins

Functions	HRG	HMWK	β_2 -GPI	CRP	SAP	C1q	MBL	TSP-1
Heparin binding	✓	✓	✓	?	✓	✓	\$	✓
Phospholipid binding	✓	✓	✓	✓	✓	✓	✓	✓
DNA/nuclear structure binding	✓	?	✓	✓	✓	✓	✓	?
Autoantibody binding	✓	✓	✓	✓	✓	✓	✓	?
Antimicrobial activity	✓	✓	✓	✓	✓	✓	✓	✓
Dying/dead cell clearance	✓	?	✓	✓	✓	✓	✓	✓
Regulates angiogenesis	✓	✓	✓	✓	✓	✓	?	✓
Regulated by plasmin cleavage	✓	✓	✓	#	#	#	#	✓
Regulates coagulation/fibrinolysis	✓	✓	✓	✓	✓	✓	✓	✓
Phosphorylated ligand binding	✓	?	?	✓	✓	?	?	?
Regulated by divalent ions	Zinc	Zinc	?	Calcium	Calcium	Calcium	Calcium	Calcium

TSP-1 is present in the plasma at a relative low concentration and is secreted by fibroblasts, monocytes, and activated platelets.

✓ indicates a function that has been reported in the literature; #, regulated by other proteases (not plasmin); \$, binds to chondroitin sulfate; β_2 -GPI, β_2 glycoprotein I; CRP, C-reactive protein; HMWK, high-molecular-weight kininogen; HRG, histidine-rich glycoprotein; MBL, mannose binding lectin; SAP, serum amyloid protein; and TSP-1, thrombospondin-1.

HRG to regulate angiogenesis, cell adhesion, cell proliferation, and remodeling of the ECM suggests that the level of endogenous HRG will influence tumor progression. Indeed, recent *in vivo* studies in which the authors used a number of murine tumor models have demonstrated the ability of HRG to inhibit the growth and vascularization of fibrosarcoma tumors,⁴⁷ as well as the development of malignant glioma (Table 1).⁴⁹ Furthermore, HRG^{-/-} mice were shown to exhibit enhanced tumor angiogenesis when the Rip1-Tag2 pancreatic tumor model was used, indicating that endogenous HRG can function as an antiangiogenic factor.⁴⁸ Contrary to these findings, studies by Klenotic et al¹⁷ demonstrated that the expression of HRG by glioma cells in both subcutaneous and orthotopic brain tumor models resulted in an increase in tumor size and angiogenesis, possibly by blocking the antiangiogenic activity of vasculostatin (Table 1).

Similarities between HRG and other functionally related plasma proteins

Plasma proteins are essential for a variety of normal physiologic processes, such as coagulation, fibrinolysis, angiogenesis, tissue remodeling, transport of nutrients, as well as the detection and clearance of unwanted materials such as pathogens, ICs, and dying/dead cells. Interestingly, several well-characterized multifunctional proteins in the plasma involved in innate immunity and tissue repair, including HMWK, β_2 glycoprotein I (β_2 -GPI), CRP, serum amyloid protein, C1q, mannose binding lectin, and TSP-1, appear to share many of the same functional properties as HRG. Although it is not surprising that proteins belonging to the same superfamily have similar biologic properties, it is becoming apparent from recent literature that HRG shares many functional similarities with other structurally unrelated plasma proteins (Table 2).

Although HRG and other functionally related serum proteins listed in Table 2 do not share any significant sequence homology (unless they are from the same superfamily), most of these proteins were reported to bind to the same endogenous ligands. The common ligands included negatively charged molecules like heparin, anionic phospholipids, and DNA. Whether these serum proteins share a similar protein motif or simply possess positively charged regions that mediate ligand binding requires further investigation. Nevertheless, their ability to interact with negatively charged molecules may facilitate their interaction with various pathogens, such as *Escherichia coli*,⁷³ *Candida albicans*,⁷⁴ and the

influenza virus,⁷⁵ which have been shown to possess negatively charged surfaces. Of importance, the ability of multiple different plasma proteins to recognize a broad range of negatively charged molecules on pathogens may provide an effective mean of clearing pathogens via different molecular mechanisms (eg, complement- vs noncomplement-based mechanisms) and thus prevent immune evasion by pathogens.

Furthermore, the ability of these functionally related proteins to interact with various phospholipids may play a vital role in dying/dead cell clearance, whereby the exposure of anionic phospholipids is a common characteristic of most apoptotic and necrotic cells.³² Indeed, most of the proteins listed in Table 2 have been reported to aid the removal of dying/dead cells.³² Interestingly, the ability of β_2 -GPI to bind to various phospholipids, especially to phospholipids exposed on dying/dead cells, has been proposed to trigger the generation of autoantibodies to β_2 -GPI and β_2 -GPI-phospholipid complexes.⁷⁶ Therefore, the production of autoantibodies against plasma proteins such as HMWK,⁷⁷ CRP,⁷⁸ and HRG^{22,79} could be initiated by the interaction between these functionally related proteins and phospholipids exposed on dying/dead cells. It would be of great interest in future studies to examine whether immunization of HRG^{-/-} mice with HRG-coated necrotic cells or simply challenging wild-type mice with necrotic cells could induce the production of autoantibodies against HRG.

Although the presence of autoantibodies against various plasma proteins can potentially lead to the onset and maintenance of pathologic conditions (eg, by blocking the normal function of the targeted protein, generating excess amounts of ICs in the circulation, or depositing autoantibodies on healthy tissues), tethering antibodies to pathogens via various opsonins may be beneficial during infections. Of interest, IgG autoantibodies against C1q,⁸⁰ β_2 -GPI,⁸¹ and possibly HRG²² often are skewed toward the IgG2 subclass, the IgG subclass that is associated with “natural” antibodies that recognize carbohydrate structures on encapsulated bacteria.⁸² Thus, the antibodies that recognize various plasma opsonins may represent “natural” antibodies, which could provide an evolutionary advantage by forming a complex with plasma opsonins that detects a broad range of molecular patterns exposed on pathogens or dying/dead cells and subsequently aids their removal via an Fc γ R-dependent mechanism. Indeed, Poon et al²² have demonstrated that the formation of HRG-IgG complexes is necessary to mediate necrotic cell removal.

Phospholipid asymmetry on the plasma membrane of dying/dead cells plays an important role in exposing “eat-me” signals,

such as phosphatidylserine, to trigger phagocytic uptake.³² Recent studies by Poon et al²² have suggested that the exposure of intracellular phospholipids like phosphatidylinositol 4-phosphate and phosphatidic acid may function as additional “eat-me” signals to enhance necrotic cell clearance via HRG. Because most of the proteins listed in Table 2 can interact with phospholipids (eg, cardiolipin, phosphatidylethanolamine, and phosphatidylserine) and have been proposed to recognize permeabilized cells,³² it is of particular interest to investigate whether these plasma opsonins can also bind to phosphatidylinositol 4-phosphate and phosphatidic acid.

Besides simply binding negatively charged phospholipids, the ability of HRG to interact with phosphatidylinositol molecules that are phosphorylated at a specific position on the inositol ring²² suggests the potential role of the phospho-moiety in mediating HRG recognition. The ability of phosphorylated molecules to function as a novel “danger” signal to indicate the presence of permeabilized cells, is an attractive hypothesis because protein phosphorylation does not usually occur extracellularly except during specific physiologic processes such as neurite outgrowth, synapse/bone formation, and myogenic differentiation.⁸³ Thus, similar to a chemokine/cytokine gradient, the release of protein complexes containing phosphorylated proteins and kinases that may phosphorylate the local extracellular environment, can potentially generate a phosphorylation gradient that may aid the recruitment of leukocytes to the site of necrotic cell death. Because CRP⁸⁴ and serum amyloid protein⁸⁵ have also been shown to bind phosphorylated carbohydrates, possibly on the surface of pathogens, these functionally related proteins may play an important role in detecting the presence of dying/dead cells and microorganisms via the recognition of phosphorylated molecules.

In addition to pathogen and dying/dead cell recognition, exposure of negatively charged surfaces may also aid the activation of the kallikrein-kinin system (KKS), which may play an important role in regulating coagulation at sites of tissue destruction or a developing thrombus, as well as other physiologic processes, including blood pressure and flow, cell proliferation, angiogenesis, apoptosis, and inflammation.⁸⁶ The KKS consists of 3 “contact factors,” namely the 2 zymogens, factor XII and prekallikrein, and the substrate/cofactor, HMWK.⁸⁶ Although the KKS was originally described as a surface-activated coagulation system that occurs on negatively charged surfaces, the assembly and activation of the KKS has also been shown to involve other cell surface molecules, such as globular heads of complement C1q, urokinase-type plasminogen activator receptor, cytokeratin 1, and tropomyosin.⁸⁶ Interestingly, besides being identified as a member of the cystatin supergene-family,⁸⁷ HRG seems to share some striking similarities with HMWK, namely the ability of HRG to bind negatively charged molecules^{1,21,22,25} and tropomyosin,⁵⁴ as well as the sensitivity of HRG to kallikrein-mediated proteolytic cleavage.⁸⁸ Thus, whether HRG participates in the KKS and regulates various biologic processes through the KKS warrants further investigation.

The ability of proteases to regulate the function of HRG⁶⁵ and other functionally related proteins is another common feature that is shared by these plasma proteins. A combination of plasmin- and kallikrein-mediated cleavage of HMWK may aid the generation of kinins,⁸⁹ which are potent vasodilator oligopeptides that can regulate blood pressure and inflammation.⁹⁰ Proteolytic cleavage of HMWK can also generate 2-chain HMWK that inhibits angiogenesis.⁹¹ Furthermore, cleavage of β_2 GPI by plasmin⁹² and polymorphonuclear neutrophil-derived proteases⁹³ has been shown to regulate the ability of β_2 GPI to exhibit antiangiogenic and antibacterial activities. Similarly, cleavage of HRG by plasmin and other

proteases has been proposed as a potential mechanism for the generation of antiangiogenic and antimicrobial fragments.^{37,47} Thus, the activation of proteases (eg, at sites of tissue injury and remodeling) may play a vital role in controlling the multifunctional activities of these functionally related proteins. Because the proteolytic cleavage of these proteins is often found to produce fragments with antiangiogenic and antimicrobial activities, it would be of great interest to examine whether the cleaved fragments of these functionally related proteins exhibit overlapping or distinct properties.

Concluding remarks

In healthy human adults, soluble proteins of the innate immune system and proteins that maintain the homeostasis of the vascular system are often present at a relatively high concentration ($> 100 \mu\text{g/mL}$) in the plasma to provide immediate responses to foreign pathogens and vascular damage. Likewise, HRG is present in human plasma at approximately $100\text{--}150 \mu\text{g/mL}$ and has been implicated in numerous biologic functions. Interestingly, HRG was found to be a negative acute phase reactant,⁹⁴ and circulating HRG levels are significantly lower during acute inflammation⁹⁵ and in patients with systemic lupus erythematosus.⁹⁶ These observations suggest that HRG is actively involved in acute inflammation as well as chronic autoimmune diseases. Although the authors of several clinical studies have reported that elevated or reduced HRG levels in patients also are associated with various thrombotic diseases,^{97,98} families with congenital HRG deficiencies (20%–35% of normal plasma level) fail to show any apparent abnormalities in routine laboratory assays of hemostatic and immunologic function, suggesting that HRG at 20% normal plasma levels is adequate to maintain normal physiologic functions.⁹⁹ Nevertheless, recent studies in which the authors used HRG^{−/−} mice have demonstrated that HRG is an important factor for efficient clearance of fungal and bacterial infections^{38,40} and may play a minor role in regulating hemostasis.⁵⁰ In contrast, whether HRG plays a predominately proangiogenic or antiangiogenic role in vivo under normal and pathologic conditions remains uncertain because conflicting results have been reported with the use of a variety of mouse models.^{17,46–49} Furthermore, because most HRG functions have been proposed on the basis of in vitro observations in which purified native/recombinant protein, protein fragments, or synthetic peptides are used, it is imperative to validate whether HRG is truly a multifunctional protein in both immunity and vascular biology using HRG^{−/−} mice (Table 1). Of importance, whether HRG simply functions as a regulator or is a key molecule in either a nonredundant or redundant pathway remains to be determined.

As described for the first time in this review, a picture is emerging of HRG being one member of an array of structurally distinct but functionally related plasma molecules. This novel family of multifunctional plasma protein is likely to play a critical role in recognizing common molecular “danger” signals in the innate response that protects against tissue damage and pathogen invasion as well as aiding wound healing.

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Correspondence: Professor Christopher Parish, The John Curtin School of Medical Research, Bldg 131, Australian National University, Garran Rd, Acton, Canberra, 2601, Australia; e-mail: Christopher.Parish@anu.edu.au; and Mark Hulett, Department of Biochemistry, La Trobe Institute for Molecular Science, La Trobe University, Physical Sciences 4 Bldg, Kingsbury Dr, Bundoora, Melbourne, 3086, Australia; e-mail: m.hulett@latrobe.edu.au.

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